International University

School of Computer Science and Engineering



Data Mining

Lab 01 – Introduction to Weka

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1. Exploring Datasets

1.1 Dataset vote.arff

Dataset	Attributes	Values	#Instances
Relation: vote #Instances: 435 #Attributes: 17	handicapped-infants	n, y, ?	n: 168, y: 219, ?: 48
	water-project-cost-sharing	n, y, ?	n: 192, y: 195, ?: 48
	adoption-of-the-budget-resolution	n, y, ?	n: 171, y: 253, ?: 11
	physician-fee-freeze	n, y, ?	n: 247, y: 177, ?: 11
	el-salvador-aid	n, y, ?	n: 208, y: 212, ?: 15
	religious-groups-in-schools	n, y, ?	n: 152, y: 272, ?: 11
	anti-satellite-test-ban	n, y, ?	n: 182, y: 239, ?: 14
	aid-to-nicaraguan-contras	n, y, ?	n: 178, y: 242, ?: 15
	mx-missile	n, y, ?	n: 206, y: 207, ?: 22
	immigration	n, y, ?	n: 212, y: 216, ?: 7
	synfuels-corporation-cutback	n, y, ?	n: 264, y: 150, ?: 21
	education-spending	n, y, ?	n: 233, y: 171, ?: 31
	superfund-right-to-sue	n, y, ?	n: 201, y: 209, ?: 25
	crime	n, y, ?	n: 170, y: 248, ?: 17
	duty-free-exports	n, y, ?	n: 233, y: 174, ?: 28
	export-administration-act-south-africa	n, y, ?	n: 62, y: 269, ?: 104
Class	Class	democrat, republican	democrat: 267, republican: 168

Table 1.1 *Explorer table of dataset vote.arff*

Observations: All 16 voting attributes are nominal with values y (yes) and n (no), plus many missing values (?). The class attribute classifies each congressperson as either democrat (267 instances,

61.4%) or republican (168 instances, 38.6%). The attribute export-administration-act-south-africa has the most missing values (104 out of 435, ~24%).

1.2 Dataset diabetes.arff

Dataset	Attributes	Values (Range)	#Instances
Relation: pima_diabetes #Instances: 768 #Attributes: 9	preg	Min: 0, Max: 17, Mean: 3.85	768
	plas	Min: 0, Max: 199, Mean: 120.89	768
	pres	Min: 0, Max: 122, Mean: 69.11	768
	skin	Min: 0, Max: 99, Mean: 20.54	768
	insu	Min: 0, Max: 846, Mean: 79.80	768
	mass	Min: 0, Max: 67.1, Mean: 31.99	768
	pedi	Min: 0.078, Max: 2.42, Mean: 0.47	768
	age	Min: 21, Max: 81, Mean: 33.24	768
Class	class	tested_negative, tested_positive	negative: 500, positive: 268

Table 1.2 *Explorer table of dataset diabetes.arff*

Observations: All 8 feature attributes are numeric. Unlike vote.arff, there are no missing values in this dataset. The class distribution is imbalanced: 500 (65.1%) tested negative vs. 268 (34.9%) tested positive. The attributes represent medical measurements like number of pregnancies, plasma glucose concentration, blood pressure, skin thickness, insulin level, BMI, diabetes pedigree function, and age.

1.3 Dataset credit-g.arff

Dataset	Attributes	Values	#Instances
Relation: german_credit #Instances: 1000 #Attributes: 21	checking_status (nominal)	<0, 0<=X<200, >=200, no checking	274, 269, 63, 394
	duration (numeric)	Min: 4, Max: 72, Mean: 20.90	1000
	credit_history (nominal)	no credits/all paid, all paid, existing paid, delayed previously, critical/other existing credit	40, 49, 530, 88, 293

	purpose (nominal)	new car, used car, furniture/equipment, radio/tv, domestic appliance, repairs, education, vacation, retraining, business, other	234, 103, 181, 280, 12, 22, 50, 0, 9, 97, 12
	credit_amount (numeric)	Min: 250, Max: 18424, Mean: 3271.26	1000
	savings_status (nominal)	<100, 100<=X<500, 500<=X<1000, >=1000, no known savings	603, 103, 63, 48, 183
	employment (nominal)	unemployed, <1, 1<=X<4, 4<=X<7, >=7	62, 172, 339, 174, 253
	installment_commitment (numeric)	Min: 1, Max: 4, Mean: 2.97	1000
	personal_status (nominal)	male div/sep, female div/dep/mar, male single, male mar/wid, female single	50, 310, 548, 92, 0
	other_parties (nominal)	none, co applicant, guarantor	907, 41, 52
	residence_since (numeric)	Min: 1, Max: 4, Mean: 2.85	1000
	property_magnitude (nominal)	real estate, life insurance, car, no known property	282, 232, 332, 154
	age (numeric)	Min: 19, Max: 75, Mean: 35.55	1000
	other_payment_plans (nominal)	bank, stores, none	139, 47, 814
	housing (nominal)	rent, own, for free	179, 713, 108
	existing_credits (numeric)	Min: 1, Max: 4, Mean: 1.41	1000
	job (nominal)	unemp/unskilled non res, unskilled resident, skilled, high qualif/self emp/mgmt	22, 200, 630, 148
	num_dependents (numeric)	Min: 1, Max: 2, Mean: 1.16	1000
	own_telephone (nominal)	none, yes	596, 404
	foreign_worker (nominal)	yes, no	963, 37
Class	class	good, bad	good: 700, bad: 300

Table 1.3 *Explorer table of dataset credit-g.arff*

Observations: This dataset has a mix of 7 numeric and 13 nominal attributes (plus the class). The class distribution is imbalanced: 700 (70%) good credit vs. 300 (30%) bad credit. The most informative attributes for credit risk tend to be checking_status, credit_history, and duration. Note that vacation has 0 instances for purpose, and female single has 0 instances for personal_status.

1.4 Dataset students.arff (custom ARFF file)

A custom ARFF file was created to demonstrate the ARFF format:

```
@relation students

@attribute hours_study numeric
@attribute attendance {high, medium, low}
@attribute part_time_job {yes, no}
@attribute result {pass, fail}

@data
10, high, no, pass
2, low, yes, fail
6, medium, no, pass
1, low, yes, fail
8, high, yes, pass
4, medium, yes, fail
```

Dataset	Attributes	Values	#Instances
Relation: students #Instances: 6 #Attributes: 4	hours_study (numeric)	Min: 1, Max: 10, Mean: 5.17	6
	attendance (nominal)	high, medium, low	high: 2, medium: 2, low: 2
	part_time_job (nominal)	yes, no	yes: 4, no: 2
Class	result	pass, fail	pass: 3, fail: 3

Table 1.4 *Explorer table of dataset students.arff*

2. Building a Classifier

2.1 Results Table

Algorithm	Pruned/unpruned	minNumObj	No. of Leaves	Correctly Classified Instances
J48	Pruned (default)	2	103	705 (70.5%)
RandomTree	N/A	0	~537 (Size: 1073)	671 (67.1%)

Table 2.1 *Results table*

2.2 Confusion Matrix — J48 (Cross-validation)

```
=== Confusion Matrix ===
   a   b   <-- classified as
 588 112 |   a = good
 183 117 |   b = bad
```

Interpretation:

- 588 instances of class "good" were correctly classified as "good" (True Positives for good)
- 112 instances of class "good" were incorrectly classified as "bad" (False Negatives for good)
- 183 instances of class "bad" were incorrectly classified as "good" (False Positives)
- 117 instances of class "bad" were correctly classified as "bad" (True Negatives for good)
- Accuracy for "good": 588/700 = 84.0% | Accuracy for "bad": 117/300 = 39.0%
- The model is much better at predicting "good" credit than "bad" credit, likely due to class imbalance (700 vs 300).

2.3 Confusion Matrix — RandomTree (Cross-validation)

```
=== Confusion Matrix ===
   a   b   <-- classified as
 542 158 |   a = good
 171 129 |   b = bad
```

Interpretation:

- 542 good correctly predicted, 158 good misclassified as bad
- 171 bad misclassified as good, 129 bad correctly predicted
- Accuracy for "good": 542/700 = 77.4% | Accuracy for "bad": 129/300 = 43.0%

2.4 Comparison Analysis

Metric	J48 (Pruned)	RandomTree
Overall Accuracy (CV)	70.5%	67.1%
Tree Size	140 nodes	1073 nodes
Number of Leaves	103	~537
Kappa Statistic	0.2467	0.2068

Table 2.4 *Comparison analysis*

Key Observations:

1. J48 outperforms RandomTree on this dataset with 70.5% vs. 67.1% accuracy (cross-validation).
2. J48 produces a much smaller, more interpretable tree (140 nodes vs. 1073 nodes) due to its pruning mechanism.
3. RandomTree achieves 100% accuracy on training data (overfitting), but this does not generalize — its cross-validation accuracy drops to 67.1%.
4. J48's pruning strategy successfully prevents overfitting: training accuracy is 85.5%, cross-validation accuracy is 70.5% — a much smaller gap than RandomTree's 100% vs. 67.1%.
5. Both classifiers struggle with predicting the "bad" class due to class imbalance (only 30% are "bad").

3. Using a Filter

Comparing J48 classification on credit-g.arff with different filter configurations (10-fold cross-validation):

Filter	No. of Leaves	Correctly Classified Instances	Remark
Original (all 20 attributes)	103	705 (70.5%)	Baseline — uses all features
Remove checking_status (19 attributes)	54	685 (68.5%)	Accuracy decreased by 2%. checking_status was an important feature for the classifier.

Keep only duration and credit_amount (2 attributes)	2	687 (68.7%)	Only marginally worse than removing checking_status, but much simpler tree (only 2 leaves).

Table 3. *Comparison table of J48 classification on credit-g.arff with different filter configurations*

Analysis:

1. Removing checking_status hurts performance (70.5% → 68.5%). This makes sense because checking account status is one of the strongest predictors of credit risk — people with no checking account or negative balance are more likely to default.
2. Using only duration and credit_amount gives 68.7% accuracy with just 2 leaves. The tree simply splits on duration ≤ 42: loans shorter than 42 months → "good", longer loans → "bad". This extremely simple model performs almost as well as the 19-attribute model, suggesting that loan duration is a powerful single predictor.
3. More attributes don't always mean better classification. The full model (20 attributes, 103 leaves) only outperforms the 2-attribute model by ~1.8 percentage points, while being vastly more complex. This illustrates the bias-variance tradeoff: simpler models may generalize better if extra features introduce noise.

4. Visualizing Your Data

4.1 J48 Results on iris.2D.arff

The iris.2D.arff dataset has 150 instances with only 2 attributes (petallength and petalwidth) and 3 classes (Iris-setosa, Iris-versicolor, Iris-virginica), each with 50 instances.

Cross-validation results:

- Correctly Classified: 144 (96%)
- Incorrectly Classified: 6 (4%)

Confusion Matrix (Cross-validation):

```
=== Confusion Matrix ===
   a   b   c   <-- classified as
  49   1   0 |   a = Iris-setosa
   0  47   3 |   b = Iris-versicolor
   0   2  48 |   c = Iris-virginica
```

4.2 Misclassified Instances

Instance	Predicted class	Actual class
Iris-virginica	Iris-virginica	Iris-virginica
Iris-virginica	Iris-virginica	Iris-virginica
Iris-virginica	Iris-virginica	Iris-virginica
Iris-virginica	Iris-virginica	Iris-virginica
Iris-virginica	Iris-virginica	Iris-virginica
Iris-virginica	Iris-virginica	Iris-virginica

Table 4.2 *Misclassified instances*

Observations:

- Iris-setosa is almost perfectly separated — only 1 instance was misclassified (as Iris-versicolor). This is because setosa has distinctly smaller petal measurements.
- Most errors occur between Iris-versicolor and Iris-virginica, which overlap in their petal length and petal width ranges. 3 versicolor instances were predicted as virginica, and 2 virginica instances were predicted as versicolor.
- The overall 96% accuracy demonstrates that even with just 2 attributes (petal length and petal width), J48 can classify iris species very effectively. The decision boundary between versicolor and virginica is the main area of uncertainty.